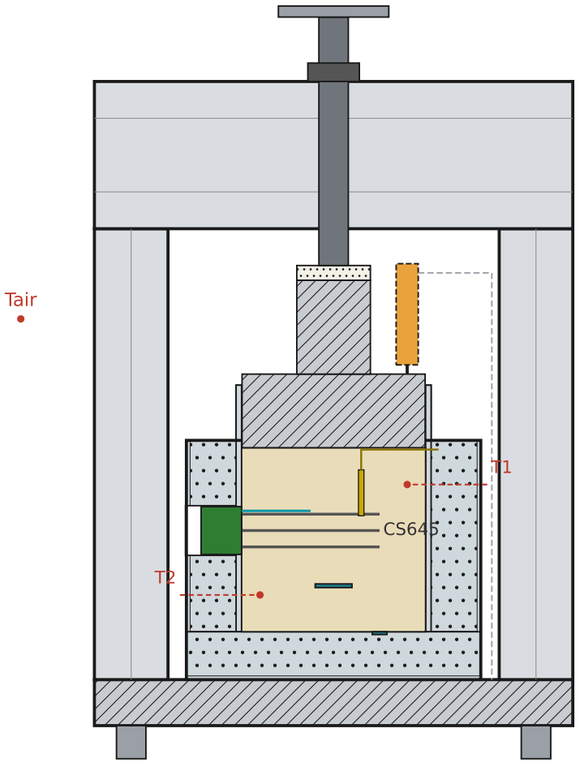
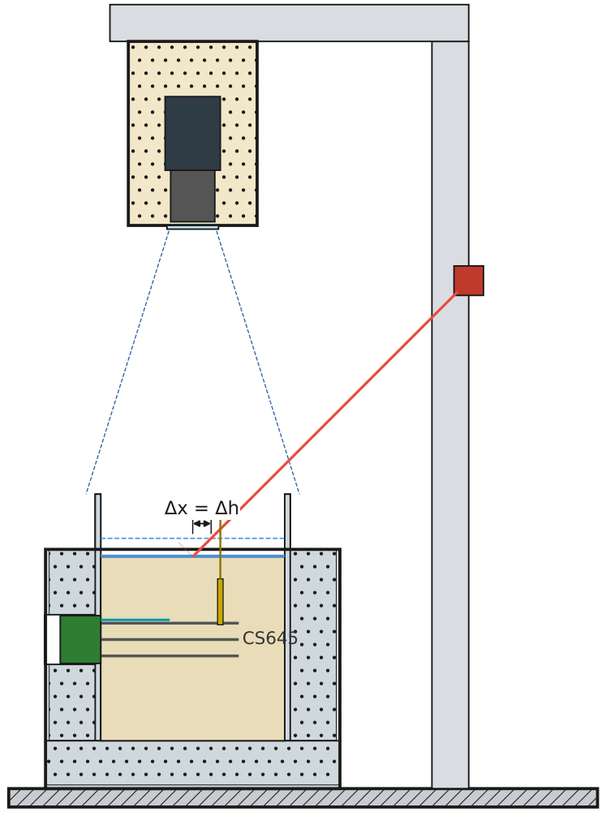
A – sample held fixed



Al frame + screw, piston locked
measures freezing PRESSURE (PV1, PB1)

B – free, open surface



nothing on the sand; camera + laser line
measure HEAVE and SETTLEMENT

Comparison

	A (V03, alt. V02)	B (V06, alt. V05)
sample	unsaturated sand ⌀100 × 100	same
cell	printed body ⌀160 (V04)	same
top of the sample	piston locked (screw)	open, nothing on it
main result	freezing pressure	heave / settlement
measured by	PV1, PB1 (TML PDA-PA)	camera + line laser
check	LVDT: piston does not move	rim = in-image reference
point M (existing)	CS645 + Pt100 + fibre	same
to buy, EUR (≈)	2 750–4 200	460–770

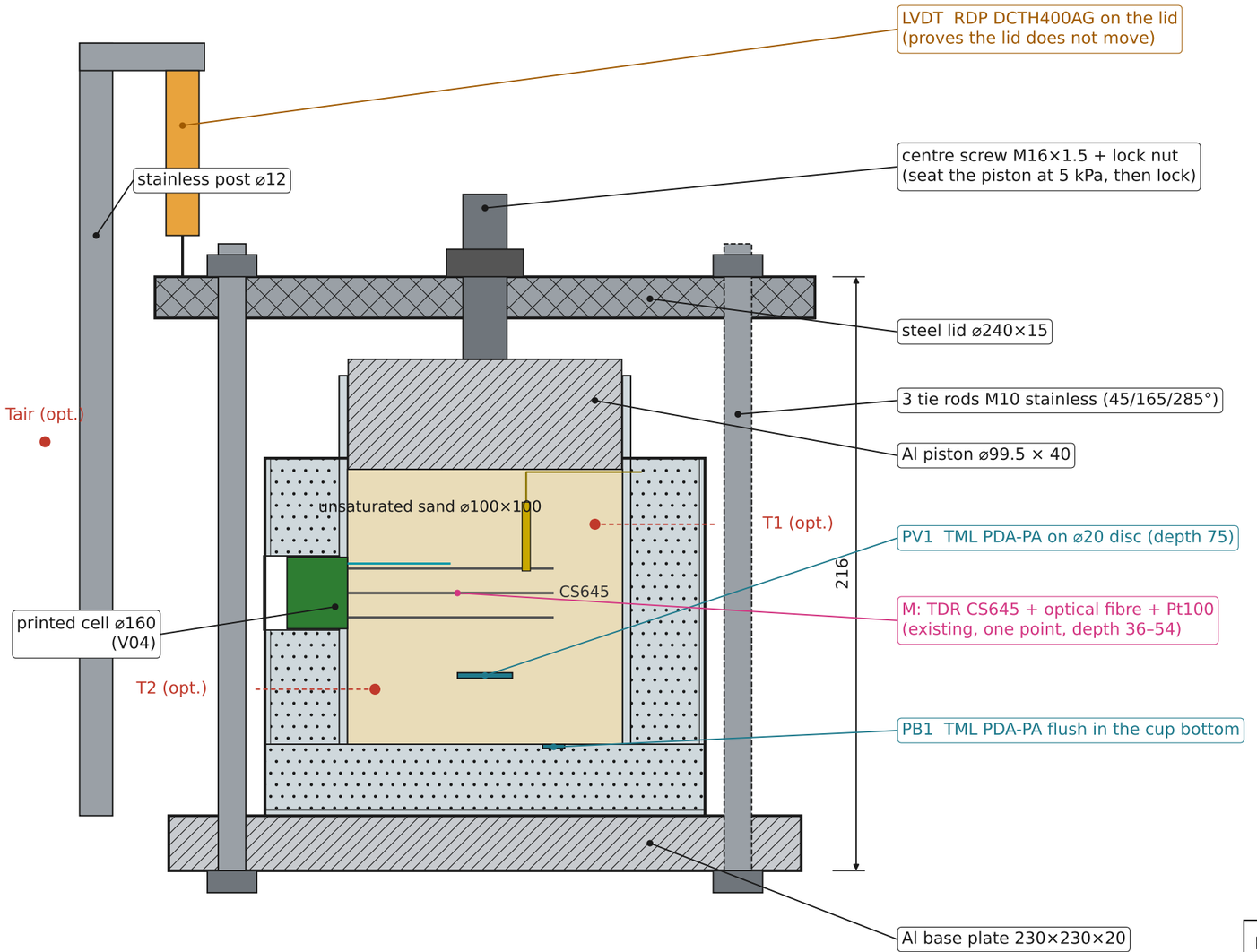
Sheets
V01 overview · V02 A with lid on tie rods · V03 A with Al frame
V04 printed cell + point M · V05 B with anchors (layer strain)
V06 B open, camera + laser · V07 sensors + CR1000 · V08 freezer

Common to both
- freezer program −10 / +10 °C, freezing from the top (insulating body)
- point M at 36–54 mm depth: unfrozen water (CS645) vs. T (Pt100, fibre)
- blank run with an Al dummy sample before the first sand run
- weigh the cell before and after each cycle

same sand, same printed cell ⌀160, same point M in both variants

E7 – overview: variant A (fixed) and B (free, open surface)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**≈ 1 : 3**	drawing no.	**E7-V01**	sheet 1 / 9



To buy – type, quantity, price (EUR excl. VAT, ≈)

tag	type	pcs	EUR	CR1000
PV1, PB1	TML PDA-PA 500 kPa (ø6.5 × 1 mm) [U]	2	1 200–2 000	DIFF1, DIFF2 BrFull, VX1
LVDT	RDP DCTH400AG (±10 mm, spring)	1	581 *	SE9
+	DIN-rail power supply for the LVDT	1	40–80	–
T1, T2, Tair	Campbell 109SS-L (ø1.6 mm)	3	540–780	SE11–13, VX3

sensors ≈ 2 361–3 441 EUR mechanics ≈ 290–590 EUR → total ≈ 2 651–4 031 EUR
* price found at a dealer (Oct 2026, see README); others = estimates → ask for quotes.
Already available (no cost): TDR CS645, Pt100, optical fibre (point M), CR1000.

Existing sensors at point M

tag	type	connection
TDR	Campbell CS645, 3 rods ø3.2 × 75, pitch 8.9	TDR100/200, SDM
Pt100	Pt100, 37 mm (25 mm metal), vertical	4-wire
OF	optical fibre, tip 37.5 from the wall	own interrogator

Fixed lid (vypocty.md §9)
• piston seated by the centre screw at 5 kPa, locked → the sample cannot heave;
  freezing pressure is read by PV1 (in the sand) and PB1 (at the bottom)
• at 1 MPa: tie rods +49 µm, lid 60 µm → piston moves ≈ 0.1 mm (≈ 20 µm at 200 kPa)
• stainless tie rods and the inner stack shrink almost equally (Δ ≈ 12 µm per 20 K)

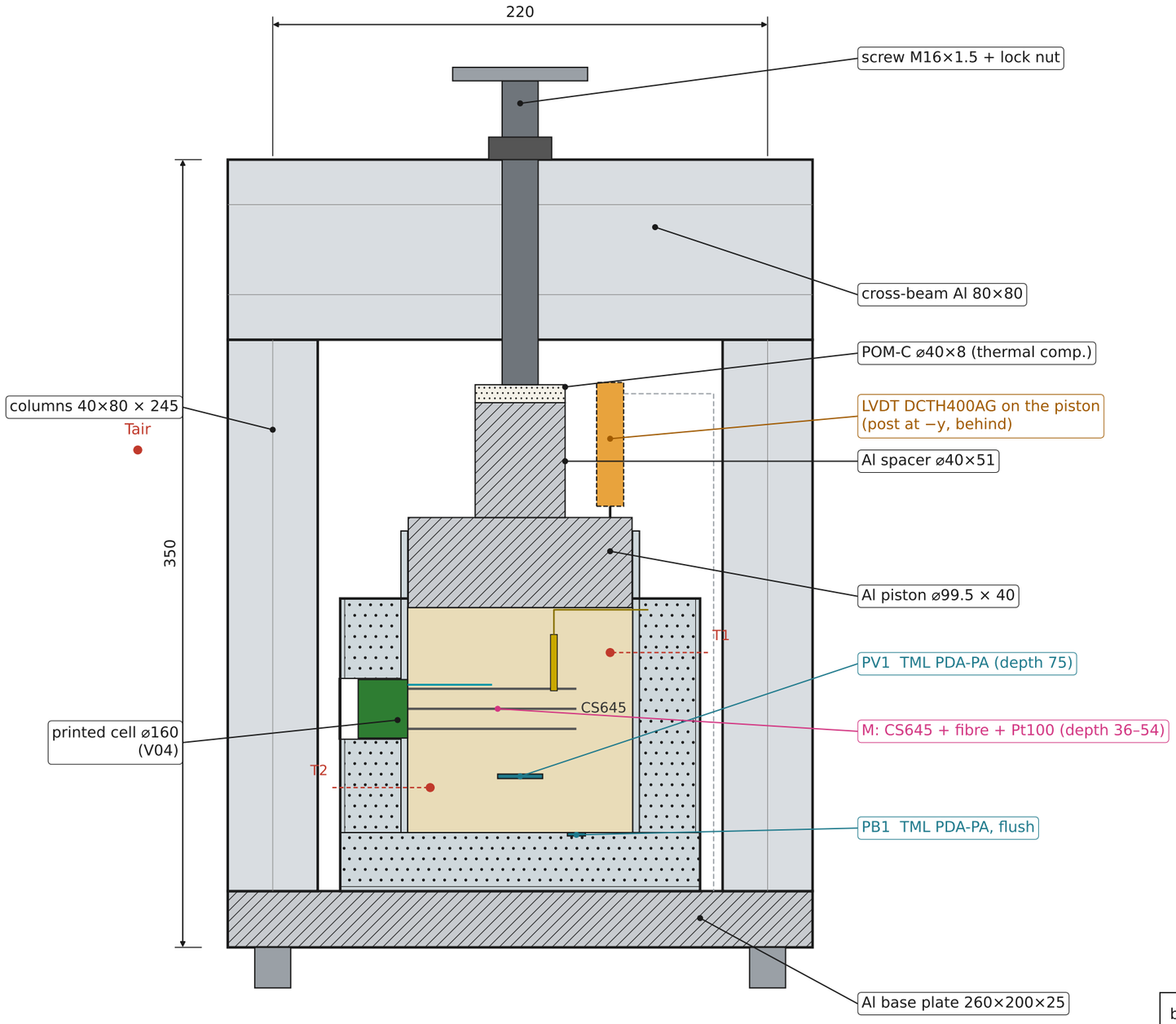
CR1000 wiring – no multiplexer
DIFF1 PV1, DIFF2 PB1 (BrFull, VX1) · SE9 LVDT · SE11–13 T1, T2, Tair (optional)
point M: per the existing devices (TDR on SDM C1–C3, temperature / fibre own reader)

Removed vs. variant A: frame, weights, carrier, ball, Invar posts, 2nd LVDT,
4 of 6 pressure cells, porous stones, pedestal, both multiplexers.

blue = pressure, red = temperature, magenta = point M

Variant A-min – fixed lid on 3 tie rods

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**≈ 1 : 2.2**	drawing no.	**E7-V02**	sheet 2 / 9



A-frame vs. A-min (tie rods)

	A-frame (this sheet)	A-min (E7-V02)
holds the piston	frame 40×80 + 80×80, screw	steel lid on 3 tie rods M10
piston movement at 1 MPa	≈ 0.08–0.15 mm	≈ 0.11 mm
thermal error (20 K)	≈ 0 with POM-C insert	≈ 12 µm
access to the sample	open sides, screw from top	lid must be removed
mechanics, EUR (≈)	≈ 375–760	≈ 330–670
sensors to buy, EUR (≈)	2 360–3 440 (same)	2 360–3 440

To buy (same sensors as A-min)

tag	type	pcs	EUR	CR1000
PV1, PB1	TML PDA-PA 500 kPa (ø6.5 × 1 mm) [U]	2	1 200–2 000	DIFF1, DIFF2 BrFull, VX1
LVDT	RDP DCTH400AG (±10 mm, spring)	1	581 *	SE9
+	DIN-rail power supply for the LVDT	1	40–80	–
T1, T2, Tair	Campbell 109SS-L (ø1.6 mm)	3	540–780	SE11–13, VX3

* price found at a dealer (Oct 2026); others estimated – ask for quotes.
Already available: TDR CS645 + reflectometer, Pt100, optical fibre (point M), CR1000.

How it works
- turn the screw until PV1 reads ≈ 5 kPa, lock the nut – the piston cannot move up;
- freezing pressure is read by PV1 (in the sand) and PB1 (bottom); LVDT proves the piston stays put (≈ 0.1 mm at 1 MPa, joints of the profiles dominate);
- Al columns shrink more than the sand stack – the POM-C insert expands the stack back (Δ ≈ 1 µm per 20 K), blank run with an Al dummy confirms it.

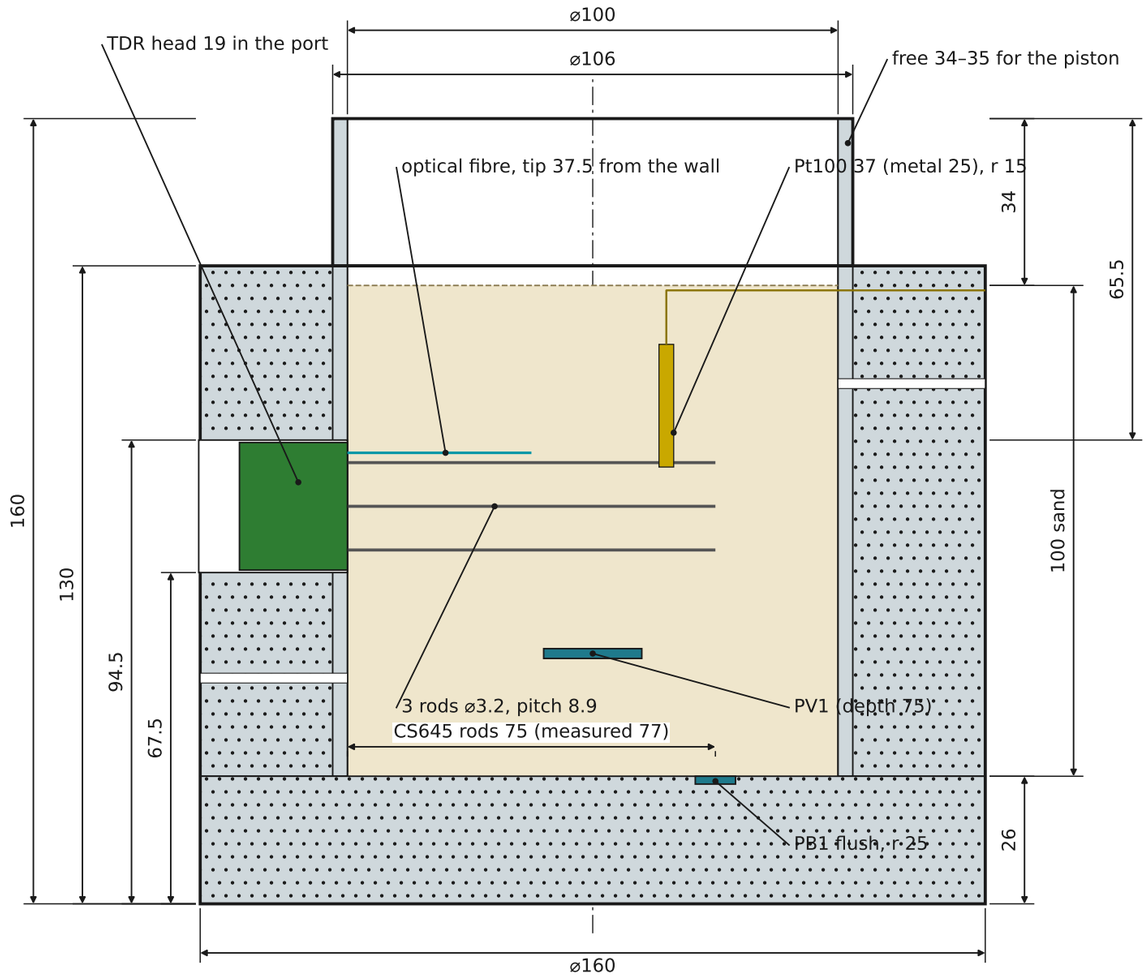
Details: vypocty.md §10.

blue = pressure, red = temperature, magenta = point M; dashed = behind

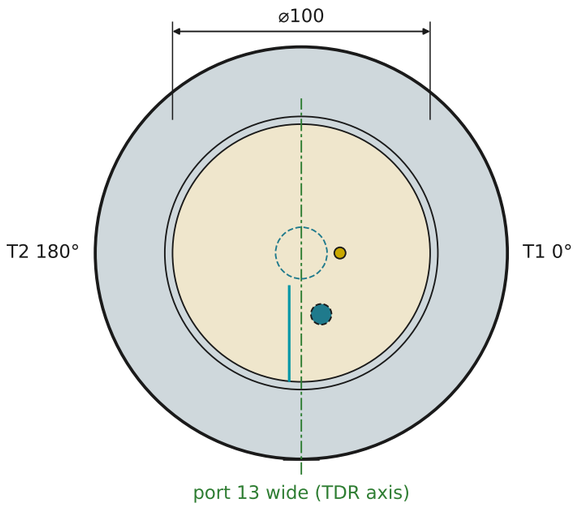
Variant A-frame – piston fixed by an Al frame

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**≈ 1 : 2.2**	drawing no.	**E7-V03**	sheet 3 / 9

SECTION A–A 1 : 1.25 (port and sensors revolved)



PLAN 1 : 2.4



Sensors at point M (existing)

	sensor	position
TDR	Campbell CS645	rods 75 (77 meas.), pitch 8.9; head fills the port, rods centred in it
OF	optical fibre	horizontal at the top rod, tip 37.5 from the wall
Pt100	37 mm (25 metal)	vertical from the sand surface, 15 from the axis

Depths below the sand surface (z = 126): rods 36 / 45 / 54,
fibre ≈ 34, Pt100 metal 12–37 → all three sense the 35–55 mm layer.
Open (sketch 7.7.26): does the Pt100 measure at the tip or along its length?

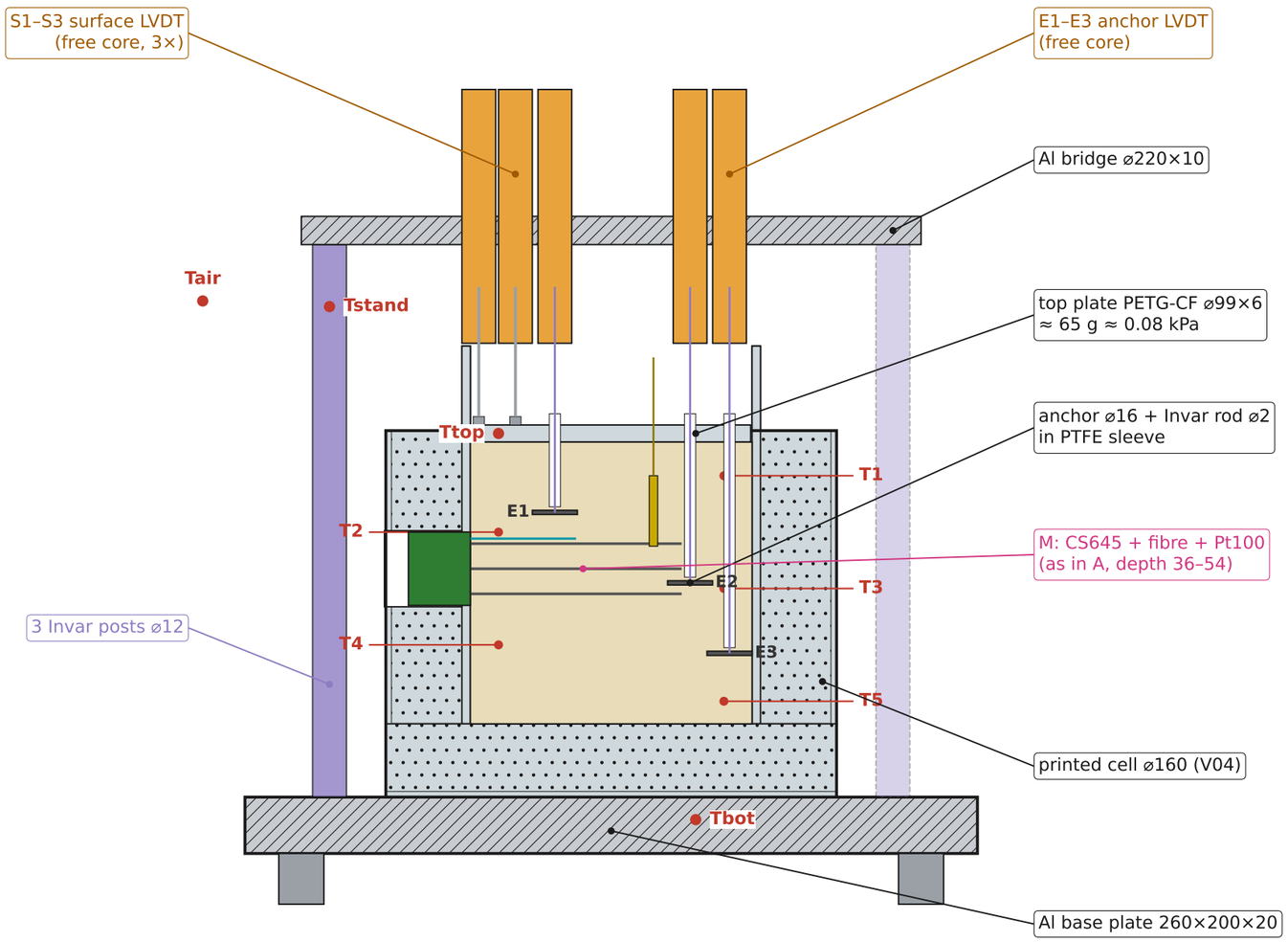
Cable routes (to print or drill)
- TDR head + coax and the fibre: through the side port 13 × 27
- Pt100 cable: groove under the piston, ⌀4 hole at z ≈ 125 (0°)
- PV1: ⌀3.5 hole at z = 51 (90°); PB1: groove in the cavity floor + ⌀3.5 hole through the jacket at 288°
- optional T1 (depth 20, 0°), T2 (depth 80, 180°): ⌀2 holes

Print (PrusaSlicer, MK4S): PETG(-CF), 0.2 mm, ≈ 600 g, 21 h
- jacket 10 % infill = insulation (replaces Kaiflex)
- recommended: 100 % infill under the cavity (z 0–26) – the bottom carries the full freezing pressure (10 % gyroid crushes at ≈ 1–2 MPa)
- wall ≈ 5 mm (≈ 8 perimeters) around the cavity: 3 mm alone sees ≈ 3 MPa hoop stress at 200 kPa, ≈ 17 MPa at 1 MPa
- epoxy coat + PTFE spray + silicone grease inside the cavity

per drawing M. Vrána 23.06.2026 + sensor sketch; z = 0 at the underside

Cell A – printed body ⌀160 (Column_one_body_version_1)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**1 : 1.25**	drawing no.	**E7-V04**	sheet 4 / 9

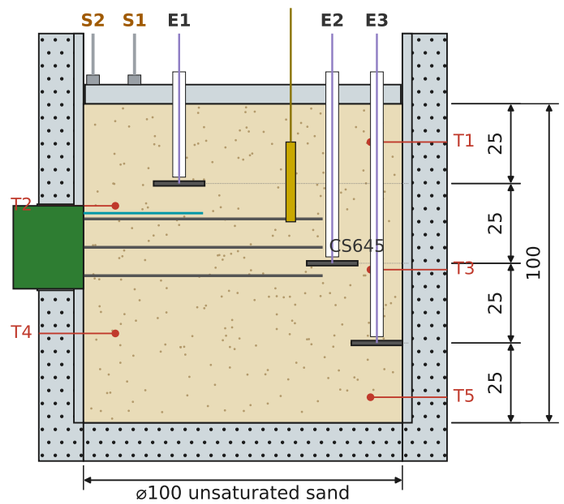


Sensors (tag → model → CR1000)

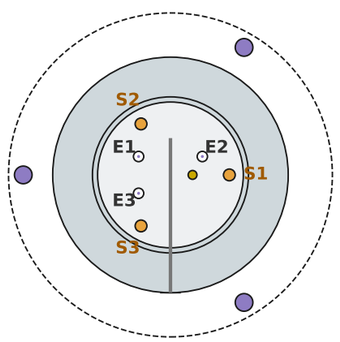
tag	measures	model	CR1000
S1…S3	surface heave	RDP DCTH200, free core [U]	SE3–5
E1…E3	anchors 25/50/75	RDP DCTH200, free core [U]	SE6–8
M: TDR	unfrozen water	Campbell CS645 (existing)	C1–C3 SDM
M: Pt100	temperature	Pt100 37 mm (existing)	4-wire
M: OF	temperature	optical fibre (existing)	own reader
T1…T5	soil, r 40	Campbell 109SS	MUX-2 SE13
Ttop, Tbot	plate, base	Campbell 109SS	(same)
Tstand, Tair	stand, air	Campbell 109SS / 107	(same)

Per freeze–thaw cycle
• heave, settlement, residual: mean of S1–S3 (difference = tilt)
• 4 layers of 25 mm: $\bar{S} - E1$, $E1 - E2$, $E2 - E3$, $E3$ (bottom fixed)
  → strain per layer, resolution ≈ $4 \cdot 10^{-5}$
• frost depth from T1–T5, unfrozen water at M
No load: free-core LVDTs push nothing, the plate adds 0.08 kPa.
Reference error ≈ 40 µm per 20 K → blank run (vypocty.md §4).

DETAIL – anchors ≈ 1 : 1.9



PLAN 1 : 4

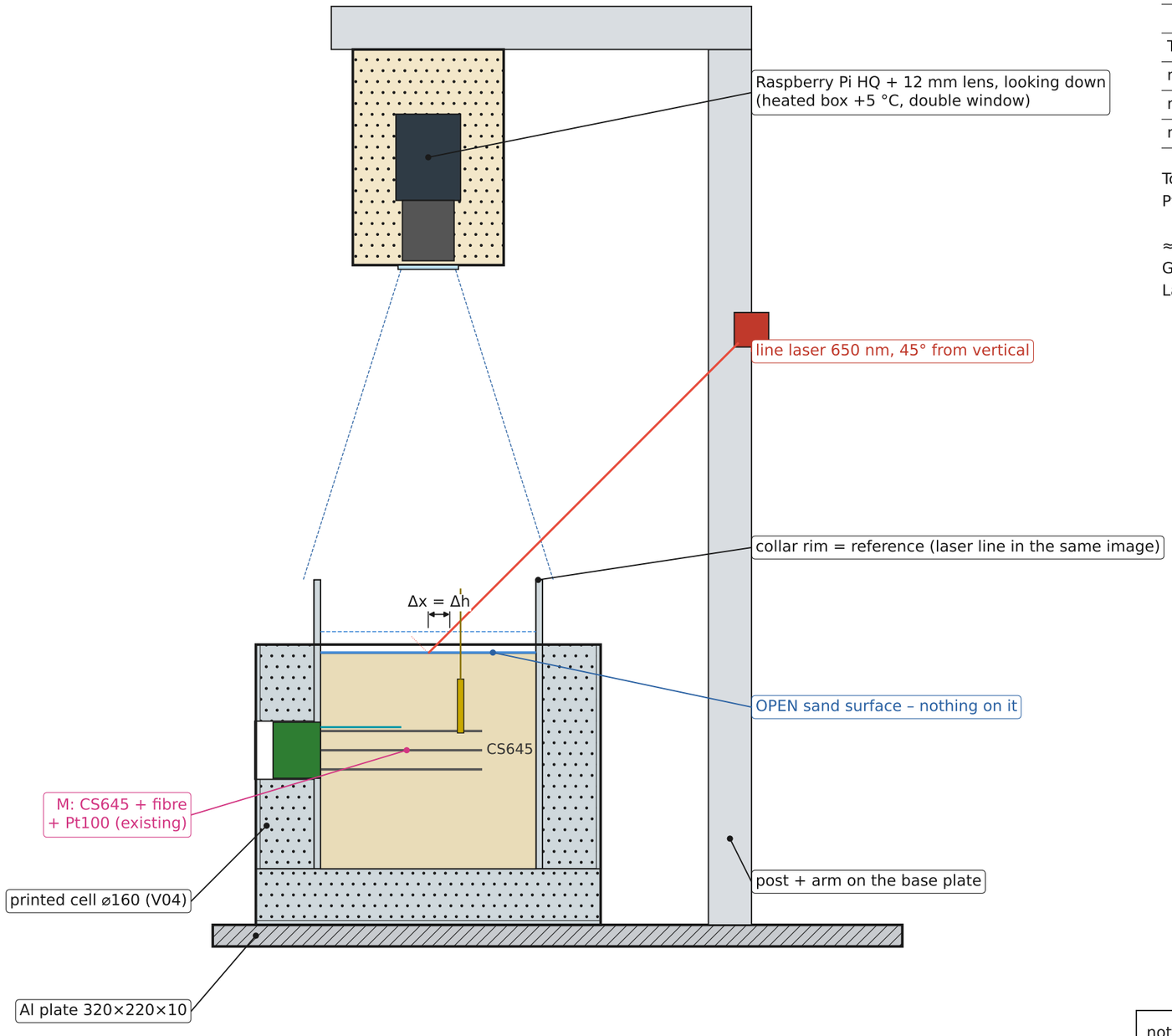


S = surface LVDT (r 40)
E = anchor (r 25)
yellow = Pt100, green = port
violet = Invar posts

orange = displacement, red = temperature, violet = Invar; revolved section

Variant B – same sample, no load: heave and layer strain

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**≈ 1 : 2**	drawing no.	**E7-V05**	sheet 5 / 9



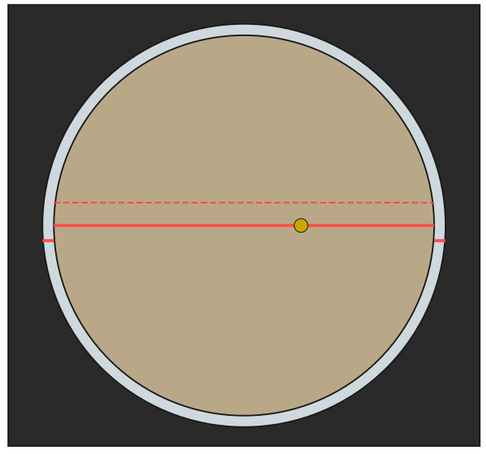
To buy (EUR excl. VAT, ≈)

tag	item	EUR
CAM	Raspberry Pi 5 (4 GB) + PSU + SD card	90–120
+	Raspberry Pi HQ Camera (IMX477, 12 MP)	55–70
+	lens 12 mm C-mount, low distortion	40–80
LAS	line laser 650 nm, class 2 (≤ 1 mW), fan 60°	15–60
+	heated box: XPS 30, heater foil 5 W, thermostat, double-glass window	40–80
Tair	Campbell 109-L (optional)	150–220
mech.	Al plate 320 × 220 × 10	40–70
mech.	post + arm over the cell (profile 20×20)	25–60
mech.	2 matt-white reference tabs on the rim (check)	2–10

Total ≈ 457–770 EUR (estimates). Already available: cell, CS645,
Pt100, optical fibre, CR1000.

≈ 28 µm/px, line centre to 0.1–0.3 px → ≈ 3–9 µm, realistic ≈ 10 µm
Gives the whole surface profile; rim moves ≈ 120 µm / 20 K → blank run.
Laser on only for the photo (every 10 min); clock synced with the CR1000.

CAMERA VIEW (from above, ≈ 115 mm field)



laser line on the rim (reference) and on the sand;
dashed = after heave → shift Δx = Δh along the whole profile

nothing on the sand: camera above, laser at 45°, rim = reference

Variant B-open – open surface, laser line + camera

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**≈ 1 : 2.5**	drawing no.	**E7-V06**	sheet 6 / 9

Sensors

rig	tag	measures	model	status	EUR
A	PV1, PB1	freezing pressure in sand / at bottom	TML PDA-PA 500 kPa [U]	buy	1 200–2 000
A	LVDT	check: piston does not move	RDP DCTH400AG ±10 mm	buy	581 + PSU
A	T1, T2	sand temp. 20 / 80 mm (optional)	Campbell 109SS-L	buy	360–520
A, B	M: TDR	unfrozen water (36–54 mm)	Campbell CS645 + TDR100/200	existing	–
A, B	M: Pt100	reference temperature	Pt100 37 mm, 4-wire	existing	–
A, B	M: OF	temperature (optical fibre)	optical fibre + reader	existing	–
B	CAM + LAS	surface profile, heave	RPi HQ camera + line laser	buy	460–770
A, B	Tair	air temperature	Campbell 109-L / 107	buy	150–220

CR1000 channel plan (no multiplexer)

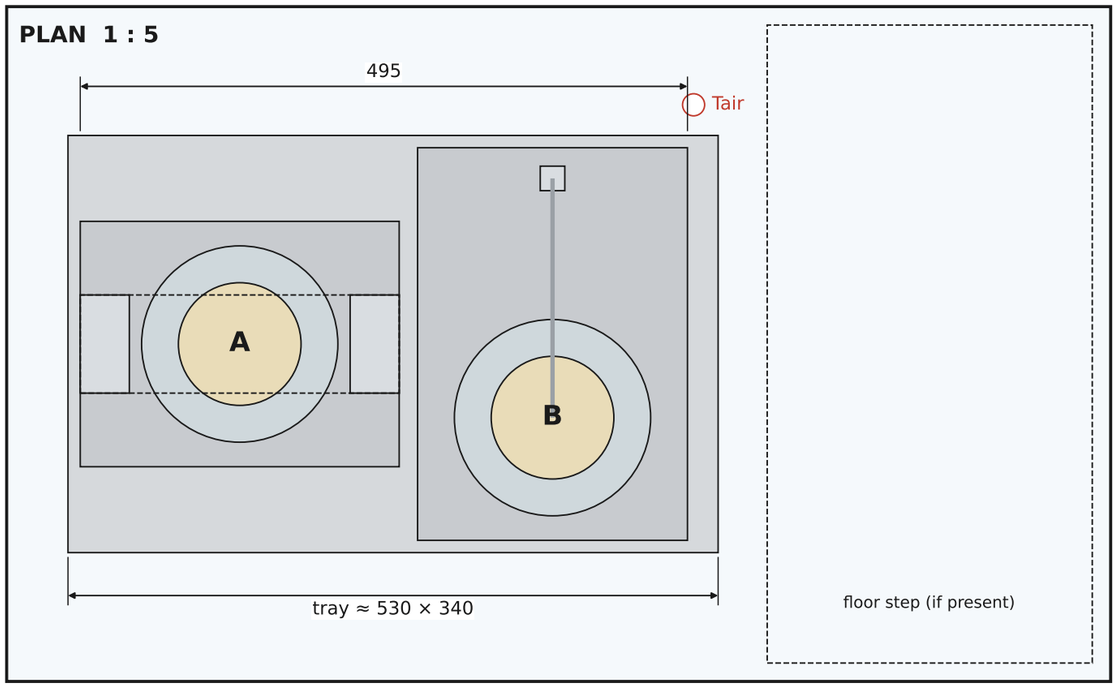
terminal	connected to	measurement
DIFF1	PV1 (A)	BrFull, VX1 2000 mV
DIFF2	PB1 (A)	BrFull, VX1 2000 mV
DIFF3	Pt100 (A) via 4WPB100	BrHalf4W, VX2 2500 mV
DIFF4	Pt100 (B) via 4WPB100	BrHalf4W, VX2 2500 mV
SE9	LVDT (A)	VoltSE ±5000 mV, own ±15 V supply
SE10, SE11	T1, T2 (A, optional)	Therm109, VX3
SE12	Tair	Therm109, VX3
C1–C3	TDR100/200 → SDMX50	2× CS645 (A and B), SDM
12V, G	TDR, LVDT supply via relay	
–	optical fibre	own interrogator, clock synced
–	camera + laser (B)	Raspberry Pi, NTP-synced, laser on per photo
SE13–16	spare	

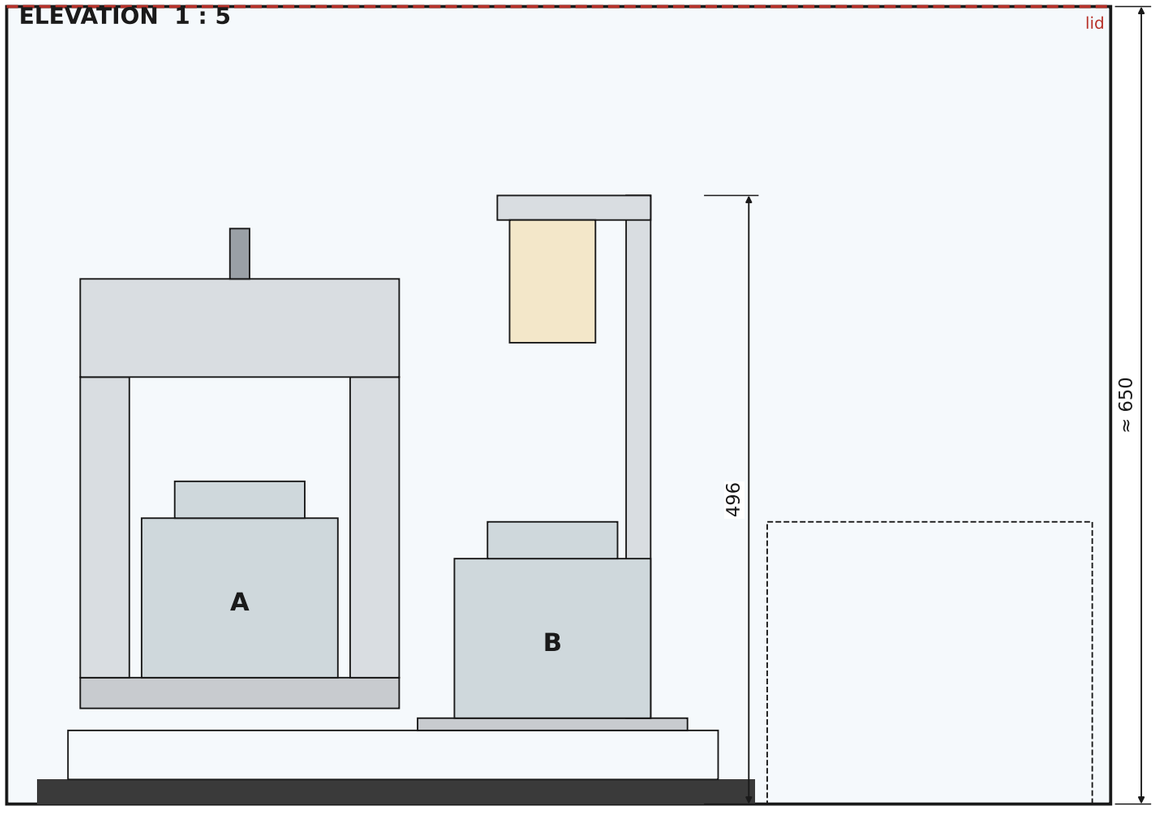
Notes
- CR1000 outside the freezer (−25…+50 °C); cables through the lid seal.
- PV1 and PB1 share VX1, the Pt100s share VX2 – the program measures them in sequence.
- CS645: one TDR instrument + SDMX50 multiplexer serves both cells. In frozen sand it gives unfrozen water; calibrate in this sand.
- Pt100 is the reference: check fibre and thermistors against it in the cell at the ice point during thawing.
- LVDT supply switched by the CR1000 (relay) – less heat in the freezer.
- Camera + laser: Raspberry Pi in a heated box, photo every 10 min, time stamps synced with the CR1000 (NTP).
- Scan interval 10 min; 1 min during the phase change.
- CR1000 is discontinued (successor CR1000X); the existing one suffices.

existing = already in the lab; prices ≈ EUR excl. VAT, ask for quotes

Sensors and CR1000 wiring (A + B-open in one freezer)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	–	drawing no.	**E7-V07**	sheet 7 / 9





Fit check

item	size	value	status
base plates	A 260×200, B 220×320	495 × 320	tray 530×340 OK
rig A height	frame + screw	409 above tray	OK
rig B height	camera arm	436 above tray	OK
highest point	above freezer floor	496	inside ≈ 650 OK
outside	CR1000, TDR, RPi	0…+40 °C	keep dry

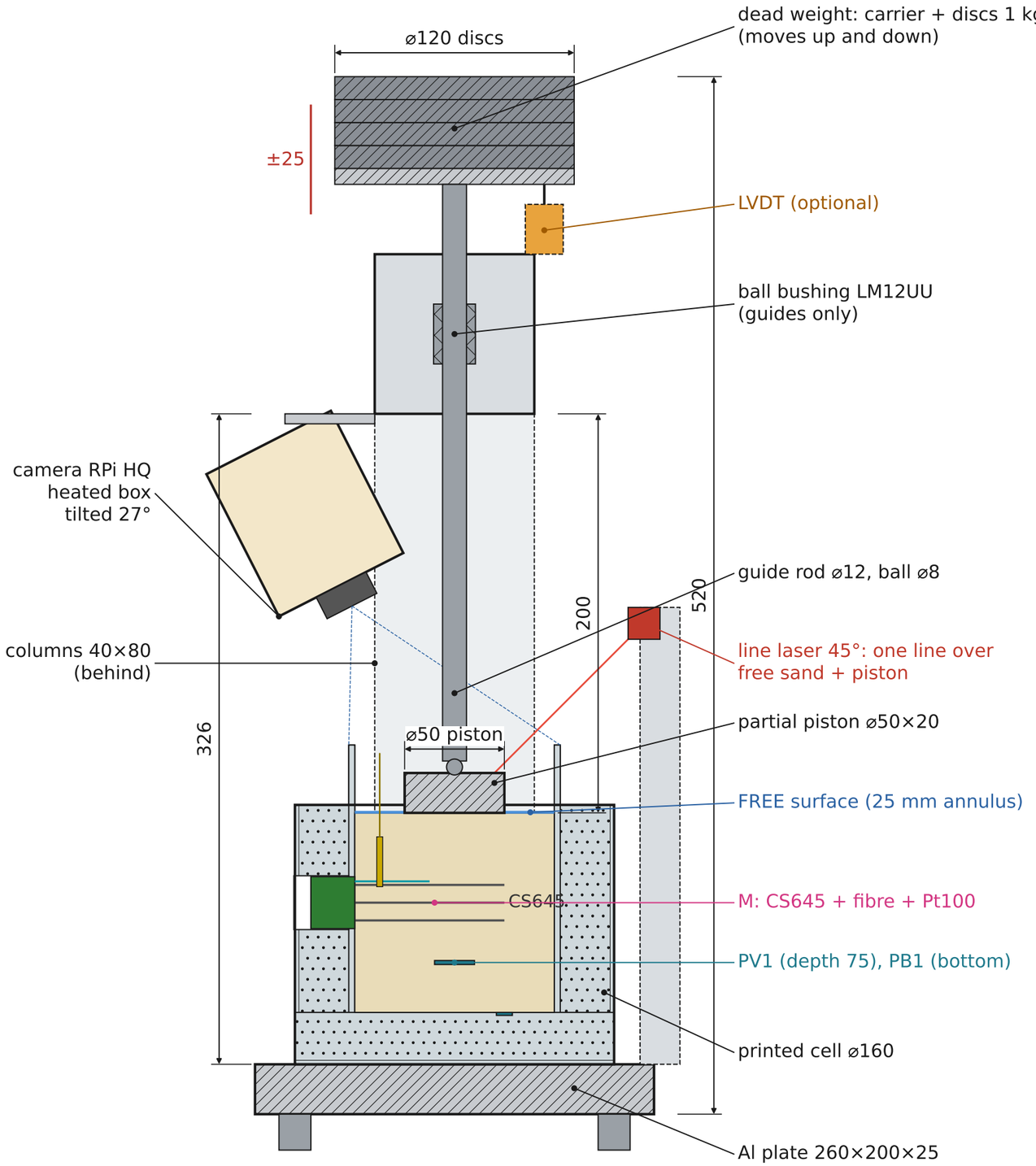
Notes

- Both rigs share the freezer program and the CR1000.
- B is turned 90° (320 mm along the tray depth) so both fit the tray.
- LED/laser of B only on for the photo; camera box ≥ 100 mm from A.
- Run the blank test with Al dummies in both cells first.
- Dimensions marked ≈ are estimates from the photos (E6) – measure.

same freezer as E6 (inside ≈ 900 × 550 × 650, to be measured); CR1000 + RPi outside

Both rigs in the chest freezer

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**1 : 5**	drawing no.	**E7-V08**	sheet 8 / 9



Load under the piston (⌀50 = 1963 mm²)

kPa	total kg	1 kg discs	trim kg
5	1.00	0	0.39
10	2.00	1	0.39
15	3.00	2	0.39
20	4.00	3	0.39
25	5.00	4	0.39

Moving part without discs ≈ 0.61 kg ≈ 3.0 kPa (minimum).

How it works

- gravity: the whole load (piston, rod, carrier, discs) rests on the sand and moves freely up (heave) and down (thaw) – constant load
- the frame and the bushing only keep the rod vertical
- partial piston ⌀50 = loaded footing; the 25 mm annulus stays FREE
- camera + laser line: heave of the loaded zone (piston top) and of the free surface in one image, collar rim = reference
- PV1 / PB1: stress under the footing; M: unfrozen water vs. T

Details: vypocty.md §14; 3D: 3d/E7_V09_variant_C_combined.step

A + B combined; side view y–z; the frame only GUIDES the load

Variant C – dead weight on a partial piston + camera

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**≈ 1 : 2.5**	drawing no.	**E7-V09**	sheet 9 / 9